An Automated Framework for Real-Time Edge-Based Leaf Disease Detection in Smart Agriculture

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ABSTRACT— Crop diseases present significant risks to global food security and agricultural yields. Traditional diagnosis relies heavily on manual inspections, which are error-prone and logistically slow. While deep learning models offer automated vision-based diagnosis, deployment on edge devices remains bottlenecked by high computational footprints and latency constraints. This paper introduces AgriEdgeNet, an optimized, lightweight convolutional neural network framework designed explicitly for localized edge-based inference. By integrating depthwise separable convolutions and an adaptive channel-attention mechanism, the proposed model reduces parameter overhead by 74% compared to standard architectures while maintaining structural classification accuracy. The framework was evaluated on an edge computing node deploying a dataset of 4,500 validation images spanning ten distinct crop pathology profiles. Experimental results demonstrate a peak classification accuracy of 98.42% with an inference latency of 42 milliseconds per frame, validating its efficacy for real-time localized deployment.

Keywords— Deep Learning, Edge Computing, Computer Vision, Smart Agriculture, Model Optimization.

I. INTRODUCTION

Modern agricultural systems are highly vulnerable to pathogen proliferation, which routinely accounts for up to 30% of aggregate annual crop yield losses globally. Early identification of localized anomalies remains the primary defensive measure against wide-scale biological contamination. Manual leaf diagnostics carried out by domain experts are inherently unscalable and highly subjective. Consequently, computer-vision frameworks driven by convolutional neural networks (CNNs) have emerged as the standard approach for automated plant pathology classification.

However, modern high-fidelity CNN models rely on heavily parameterized, centralized cloud architectures to function efficiently. In practical agricultural fields, persistent high-bandwidth network connectivity to cloud servers is frequently absent or cost-prohibitive. Transferring raw image arrays continuously from field cameras to remote datacenters introduces prohibitive latency spikes and security exposure. Therefore, localizing inference

directly onto resource-constrained edge computing architectures situated within the physical farm infrastructure represents a critical paradigm shift.

This paper presents the design, optimization, and empirical deployment evaluation of AgriEdgeNet. The contributions of this work are three-fold: (1) design of a specialized, highly compressed network topology that leverages depthwise separable convolutions to match resource constraints, (2) an integration of localized channel-attention matrices to safeguard feature retrieval integrity during downsampling, and (3) an empirical bench test assessing latency-to-accuracy dynamics across native edge hardware profiles.

II. RELATED WORK

Automated plant leaf disease classification has matured substantially due to deep learning advancements. Early approaches relied on manual color space thresholding and Support Vector Machines (SVMs). These architectures lacked robust generalization performance when tested under dynamic ambient lighting variations typical of outdoor farmlands.

Recent implementations have adopted deeper networks such as ResNet and VGG-16. While these networks yield excellent raw validation metrics, their parameter volumes exceed 20 to 130 million floating-point variables. Such structural properties limit practical execution on decentralized edge processing units due to limited static memory storage and rigid thermal limits. Recent efforts have utilized MobileNet variations; however, these generic models often omit fine-grained leaf vein patterns, dropping accuracy when encountering early-stage micro-lesions. Our proposed AgriEdgeNet fills this technical gap by embedding custom attention profiles that prioritize low-level spatial geometry without inflating parameters.

III. METHODOLOGY

A. Architecture Design

The fundamental architecture of AgriEdgeNet relies on a sequence of deep convolutional blocks engineered to scale down computational complexity. Standard convolutional layers are systematically replaced with Depthwise Separable Convolutions (DSC). Each DSC operation splits standard kernel passes into two distinct layers: a single spatial filtering matrix followed immediately by a pointwise channel-mixing matrix.

To prevent feature degradation, an adaptive channel-attention module is structurally inserted into the bottleneck layers. This block generates channel-specific weight distributions by executing an global average pooling operation over the incoming spatial dimensions. The generated attention coefficient vector scales the feature maps to emphasize relevant structural indicators, such as localized necrotic spots or subtle leaf chlorosis.

B. Mathematical Framework

The computational savings afforded by substituting standard convolutions with depthwise separable layers can be formally modeled. Let D_K denote the spatial width of the kernel, M represent the number of input channels, and N represent the count of output channels. The direct operational cost scaling factor C_R is formalized in Equation 1:

$$C_R = \frac{D_K^2 \cdot M + M \cdot N}{D_K^2 \cdot M \cdot N} = \frac{1}{N} + \frac{1}{D_K^2} \quad (1)$$

Given standard $3 imes 3$ kernels where $D_K = 3$, this mathematical substitution reduces the model's aggregate computing overhead by roughly 88% compared to standard convolution pipelines handling identical feature depths.

IV. EXPERIMENTAL RESULTS

The optimized model was compiled and executed natively on an embedded Raspberry Pi 4 compute node equipped with 4GB of low-power LPDDR4 memory. The training environment utilized a dataset containing 4,500 high-resolution leaf images distributed across ten distinct tomato and potato plant health profiles. The dataset was partitioned into an 80/20 train-to-validation matrix, running across 100 learning epochs using the Adam optimizer.

TABLE I: QUANTITATIVE COMPARATIVE FRAMEWORK PERFORMANCE

Architecture Model	Validation Accuracy (%)	Total Parameters	Latency per Frame (ms)
VGG-16 Baseline	98.11%	138.3 Million	342 ms
ResNet-50	98.65%	25.6 Million	189 ms
MobileNetV2	95.20%	3.4 Million	38 ms
AgriEdgeNet (Proposed)	**98.42%**	**1.1 Million**	**42 ms**

Table I charts the empirical comparative performance metrics across different networks. AgriEdgeNet achieves a validation accuracy profile comparable to standard cloud-reliant architectures while reducing parameter storage limits down to 1.1 million parameters. The 42 ms real-time processing window falls within acceptable tracking parameters for moving agricultural rovers or manual drone arrays.

V. CONCLUSION

This study demonstrated a lightweight, optimized deep learning framework for edge-based real-time leaf disease detection in smart agriculture environments. By combining depthwise separable convolutions and channel-attention modules, the AgriEdgeNet framework resolves the traditional tradeoff between edge-device resource constraints and semantic classification accuracy. Future field studies will expand this methodology to multi-spectral imaging arrays and test local federated learning strategies across distributed networks of edge nodes.

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